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CO2-AT2-SCENARIO BASED

COURSE CODE: DSA0308

COURSE NAME: NATURAL LANGUAGE PROCESSING

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ANSWERS:

Q1. Morphological Parsing of disagree, agreement, and agreeable

AIM

To develop a Python program that performs morphological parsing of the words disagree, agreement, and agreeable by identifying prefixes, roots, and suffixes, classifying derivational transformations, and interpreting their semantic effects for sentiment analysis.

PROBLEM STATEMENT

An AI-based sentiment analysis platform receives product reviews containing words like “disagree,” “agreement,” and “agreeable.” The system must interpret sentiment consistently despite variations in word formation. Develop a morphological parsing strategy to identify prefixes, suffixes, and base forms while capturing how derivational changes influence meaning. Write a Python program to extract roots, classify transformations, and display the original word, prefix, root, suffix, transformation category, semantic interpretation, and normalized base form.

ALGORITHM

1. Start the program.
2. Initialize the input words as disagree, agreement, and agreeable.
3. Identify the prefix, root word, and suffix using rule-based morphological analysis.
4. Classify each transformation as derivational.
5. Determine how each prefix or suffix changes the meaning or grammatical category of the root.
6. Normalize the words to the common base form agree.
7. Display the original word, prefix, root, suffix, transformation type, semantic interpretation, and normalized form in tabular format.
8. Stop the program.

1.MORPHOLOGICAL DECOMPOSITION

Input word	Prefix	Root	Suffix	Type	Semantic interpretation	Normalized form
disagree	dis-	agree	None	Derivational	Opposition or lack of agreement	agree
agreement	None	agree	-ment	Derivational	Shared opinion or understanding	agree
agreeable	None	agree	-able	Derivational	Pleasant or acceptable	agree

2. EXPLANATION

- disagree: dis- + agree — the prefix dis- expresses opposition or lack of agreement.
- agreement: agree + -ment — the suffix -ment converts the verb into a noun.
- agreeable: agree + -able — the suffix -able forms an adjective meaning pleasant or acceptable.

Step-by-step analysis

1. disagree = dis- + agree

- Prefix: dis-
- Root: agree
- Suffix: None
- Type: Derivational
- Meaning: To oppose or have a different opinion.
- Normalized form: agree.

2. agreement = agree + -ment

- Prefix: None
- Root: agree
- Suffix: -ment
- Type: Derivational
- Meaning: A shared opinion, arrangement, or understanding.
- Normalized form: agree.

3. agreeable = agree + -able

- Prefix: None

- Root: agree
- Suffix: -able
- Type: Derivational
- Meaning: Pleasant, acceptable, or willing to agree.
- Normalized form: agree.

All three words share the base agree, but their meanings differ. A sentiment analysis system should preserve the negative or opposing meaning of disagree rather than treating all three words as sentimentally identical.

3. SEMANTIC ANALYSIS

Morphological parsing identifies how prefixes and suffixes modify the meaning and grammatical category of a root word.

Word	Morphological change	Word class	Semantic effect
disagree	dis- + agree	Verb	Expresses opposition
agreement	agree + -ment	Noun	Represents a shared opinion
agreeable	agree + -able	Adjective	Describes something pleasant or acceptable

4. APPLICATION IN SENTIMENT ANALYSIS

- Negation detection: The prefix dis- helps identify opposing opinions in product reviews.
- Feature normalization: Related words can be grouped under the common root agree.
- Sentiment preservation: The original word must be retained to distinguish disagreement from agreement.
- Opinion classification: Semantic interpretation helps classify customer feedback as positive, negative, or neutral, depending on context.

Example:

Customer review	Important word	Possible interpretation
I disagree with the product quality.	disagree	Negative opinion
We reached an agreement with the seller.	agreement	Agreement; sentiment depends on context
The product is agreeable to use.	agreeable	Potentially positive

CODE

```
#jothika-192324235
words = ["disagree", "agreement", "agreeable"]
data = {
    "disagree": ( "dis-", "agree", "None", "Derivational",
        "Opposition or lack of agreement" ),
    "agreement": ("None", "agree", "-ment", "Derivational",
        "Shared opinion or understanding" ),
    "agreeable": ("None", "agree", "-able", "Derivational",
        "Pleasant or acceptable" )
}
print(f"{'Word':<13}{'Prefix':<10}{'Root':<10}{'Suffix':<10}"
      f"{'Type':<16}{'Meaning':<38}{'Normalized'}")
print("-" * 110)
for word in words:
    prefix, root, suffix, kind, meaning = data[word]
 print(f"{word:<13}{prefix:<10}{root:<10}{suffix:<10}"
          f"{kind:<16}{meaning:<38}{root}")
```

EXPECTED OUTPUT

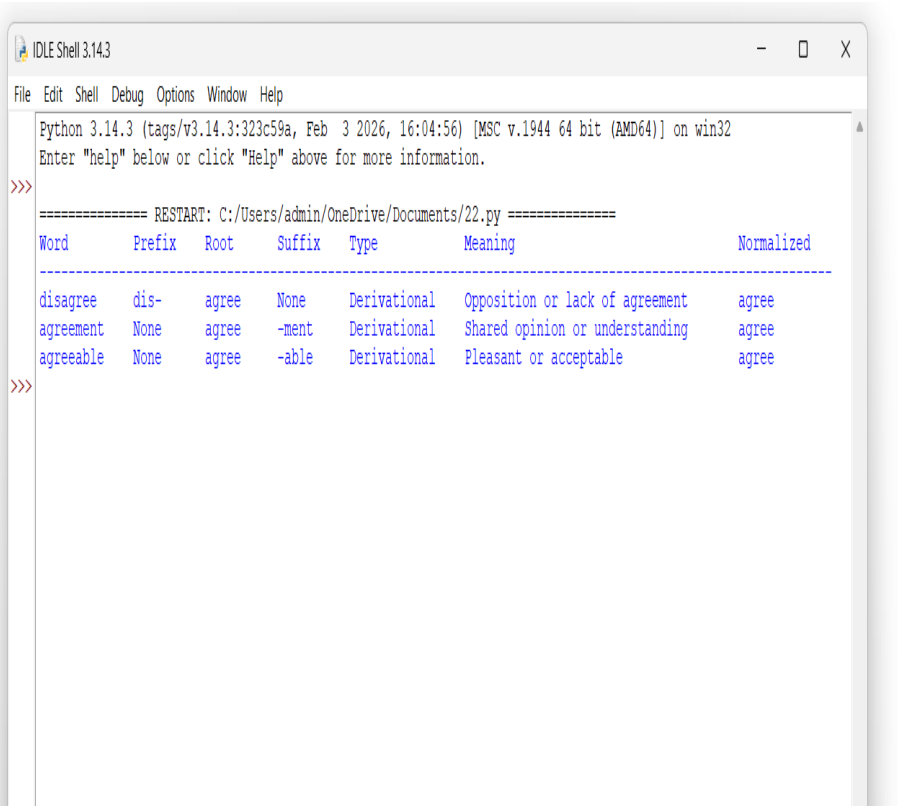
```
Word         Prefix    Root      Suffix    Type             Meaning                              Normalized
--------------------------------------------------------------------------------------------------------------
disagree     dis-      agree     None      Derivational     Opposition or lack of agreement      agree
agreement    None      agree     -ment     Derivational     Shared opinion or understanding      agree
agreeable    None      agree     -able     Derivational     Pleasant or acceptable               agree
```

OUTPUT

```
words = ["disagree", "agreement", "agreeable"]
data = {
    "disagree": (
        "dis-", "agree", "None", "Derivational",
        "Opposition or lack of agreement"
    ),
    "agreement": (
        "None", "agree", "-ment", "Derivational",
        "Shared opinion or understanding"
    ),
    "agreeable": (
        "None", "agree", "-able", "Derivational",
        "Pleasant or acceptable"
    )
}
print(f"{'Word':<13}{'Prefix':<10}{'Root':<10}{'Suffix':<10}"
      f"{'Type':<16}{'Meaning':<38}{'Normalized'}")
print("-" * 110)

for word in words:
    prefix, root, suffix, kind, meaning = data[word]

    print(f"{word:<13}{prefix:<10}{root:<10}{suffix:<10}"
          f"{kind:<16}{meaning:<38}{root}")
```



Conclusion: Thus, the Python program successfully identifies the prefixes, roots, and suffixes of disagree, agreement, and agreeable, classifies their derivational transformations, and normalizes them to agree while preserving their semantic differences for sentiment analysis.

Q2.Morphology-Based Normalization for Topic Modeling and Clustering

AIM

To develop a Python program that performs morphology-based normalization of govern, government, and governance by identifying roots, derivational suffixes, word classes, and derivational levels, and mapping related terms to a common representation for topic modeling and document clustering.

PROBLEM STATEMENT

A text analytics pipeline in a news aggregation system processes documents containing terms such as “govern,” “government,” and “governance.” The system must standardize these variations for topic modeling and clustering. Develop a morphology-based normalization mechanism that identifies roots, derivational suffixes, and derivational layers, maps the words to a common base form, and displays the original word, root, affixes, derivational hierarchy, normalized representation, and final output using Python.

ALGORITHM

1. Start the program.
2. Initialize the input words as govern, government, and governance.
3. Identify the root word and derivational suffix of each input.
4. Determine the word class and derivational level.
5. Assign level 0 to the base word and level 1 to words formed using one derivational suffix.
6. Normalize all three words to the common base govern.
7. Display the original word, root, affixes, derivational hierarchy, level, and normalized representation in tabular format.
8. Stop the program.

1.MORPHOLOGICAL DECOMPOSITION

Input word	Root	Affix	Word class	Derivational level	Normalized form
govern	govern	None	Verb	0	govern
government	govern	-ment	Noun	1	govern
governance	govern	-ance	Noun	1	govern

2. EXPLANATION

- govern: The base verb meaning to rule or manage.
- government: govern + -ment — the suffix -ment forms a noun referring to a governing body or system.
- governance: govern + -ance — the suffix -ance forms a noun referring to the process or system of governing.

Step-by-step analysis

1. govern

- Root: govern
- Affix: None
- Word class: Verb
- Derivational level: 0
- Normalized form: govern.

2. government

- Root: govern
- Suffix: -ment
- Word class: Noun
- Derivational hierarchy: govern → government
- Derivational level: 1
- Normalized form: govern.

3. governance

- Root: govern
- Suffix: -ance
- Word class: Noun
- Derivational hierarchy: govern → governance
- Derivational level: 1
- Normalized form: govern.

All three words share the lexical base govern. The derivational level indicates the number of suffix-forming steps represented in this simplified analysis.

3. APPLICATION IN TOPIC MODELING AND CLUSTERING

- Vocabulary reduction: Related word forms can be mapped to a common representation.
- Document clustering: Articles discussing governing activities can be grouped more consistently.
- Topic modeling: Normalization can help identify common themes across news articles.
- Information retrieval: A search for govern can be expanded to find documents containing government and governance.
- Lexical standardization: Different forms are processed consistently during text preprocessing.

Example:

News sentence	Identified word	Normalized form
Leaders govern the region.	govern	govern
The government announced a policy.	government	govern
Good governance improves transparency.	governance	govern

4. ADVANTAGES

1. Reduces the number of distinct vocabulary items.
2. Improves consistency in document indexing.
3. Helps group related news articles.
4. Supports common-term extraction for topic modeling.
5. Improves retrieval of documents containing morphological variants.

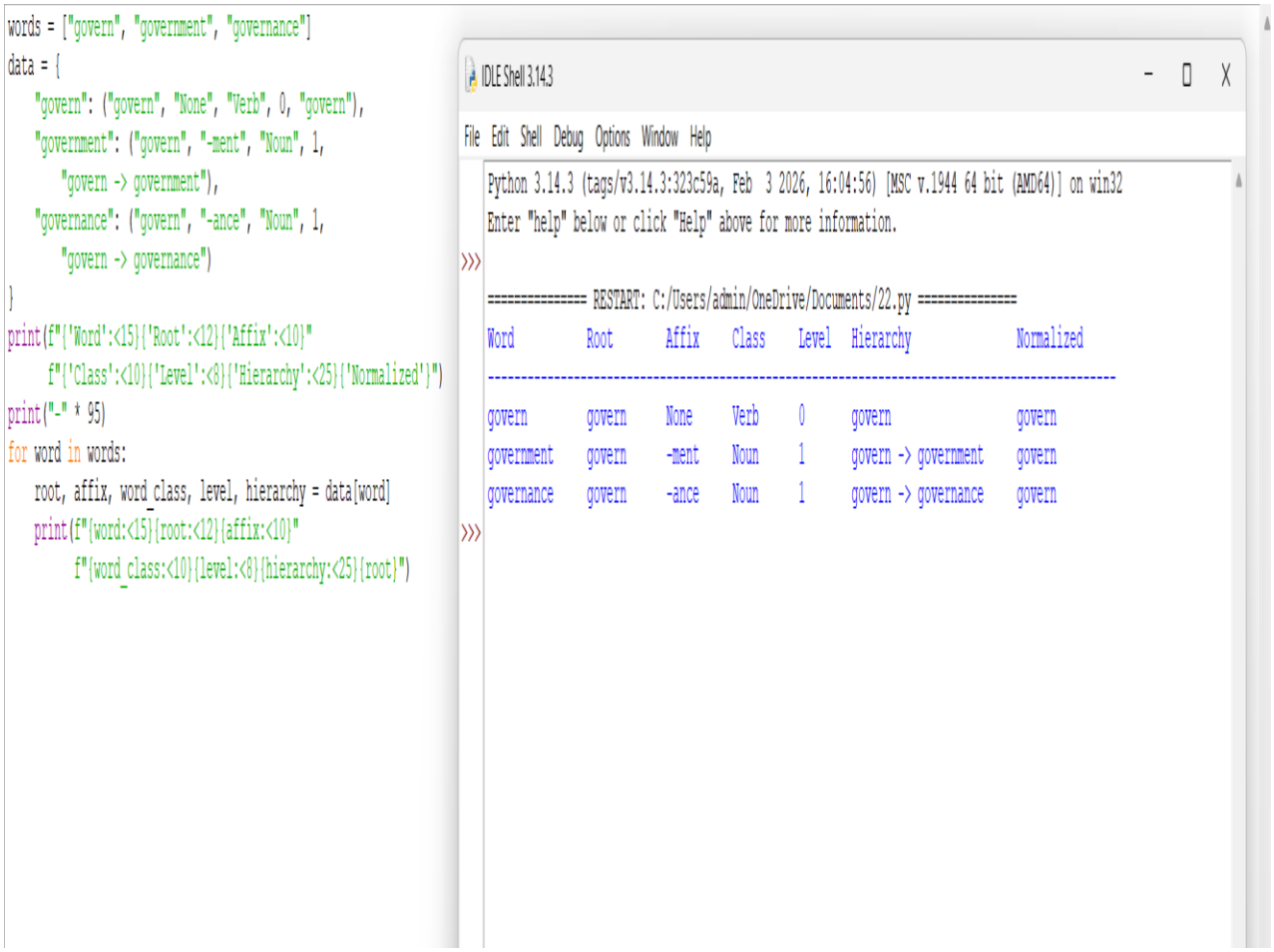
CODE

```
#jothika-192324235
words = ["govern", "government", "governance"]
data = {
    "govern": ("govern", "None", "Verb", 0, "govern"),
    "government": ("govern", "-ment", "Noun", 1,
        "govern -> government"),
    "governance": ("govern", "-ance", "Noun", 1,
        "govern -> governance")
}
print(f"{'Word':<15}{'Root':<12}{'Affix':<10}"
     f"{'Class':<10}{'Level':<8}{'Hierarchy':<25}{'Normalized'}")
print("-" * 95)
for word in words:
    root, affix, word_class, level, hierarchy = data[word]
    print(f"{word:<15}{root:<12}{affix:<10}"
          f"{word_class:<10}{level:<8}{hierarchy:<25}{root}")
```

EXPECTED OUTPUT:

```
Word           Root        Affix      Class      Level   Hierarchy                 Normalized
-----------------------------------------------------------------------------------------------
govern         govern      None       Verb       0       govern                    govern
government     govern      -ment      Noun       1       govern -> government      govern
governance     govern      -ance      Noun       1       govern -> governance      govern
```

OUTPUT



```
words = ["govern", "government", "governance"]
data = {
    "govern": ("govern", "None", "Verb", 0, "govern"),
    "government": ("govern", "-ment", "Noun", 1,
        "govern -> government"),
    "governance": ("govern", "-ance", "Noun", 1,
        "govern -> governance")
}
print(f"{'Word':<15}{'Root':<12}{'Affix':<10}"
      f"{'Class':<10}{'Level':<8}{'Hierarchy':<25}{'Normalized'}")
print("-" * 95)
for word in words:
    root, affix, word_class, level, hierarchy = data[word]
    print(f"{word:<15}{root:<12}{affix:<10}"
          f"{word_class:<10}{level:<8}{hierarchy:<25}{root}")
```

```
IDLE Shell 3.14.3
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/admin/OneDrive/Documents/22.py ==========
Word           Root        Affix      Class      Level   Hierarchy                 Normalized
---------------------------------------------------------------------------------------------
govern         govern      None       Verb       0       govern                    govern
government     govern      -ment      Noun       1       govern -> government      govern
governance     govern      -ance      Noun       1       govern -> governance      govern
>>>
```

Conclusion: All three words share the normalized root happy. However, unhappy expresses negative sentiment, so a sentiment analysis system should preserve the negation instead of treating all three words as having identical sentiment.